

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (CSE) (CBCS) Degree Examinations

December 2023

CS2101-1 – COMPUTER ORGANIZATION & ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following registers stores the address of memory location to be accessed.
A) MAR B) MDR
C) IR D) PC
- is the processor performance equation, where E_T is the processor time required to execute a program N is the actual number of instructions executed, S is the average number of basic steps needed to execute one machine instruction, and R is the clock rate.
A) $E_T = (N \times R) / S$ B) $E_T = (R \times S) / N$
C) $E_T = (N \times S) / R$ D) $E_T = S / (N \times R)$
- The overall SPEC (System Performance Evaluation Corporation) ratio for a computer is given by, where n is the number of programs in the SPEC suite and $SPEC_i$ is the ratio for the program _{i} in the suite.
A) $(\prod_{i=1}^n SPEC_i)^{\frac{1}{n}}$ B) $(\sum_{i=1}^n SPEC_i)^{\frac{1}{n}}$
C) $\frac{\text{running time on the reference computer}}{\text{running time on the computer under test}}$ D) $\frac{\text{running time on the computer under test}}{\text{running time on the reference computer}}$
- In addressing mode Effective Address is calculated as $EA = [R_i]$
A) direct B) indirect
C) indexed D) immediate
- According to Booth's algorithm, the multiplier code for 10001 is
A) -1 0 0 +1 -1 B) +1 0 0 +1 +1
C) +1 0 0 +1 -1 D) -1 0 0 -1 +1
- The number of bits in the mantissa field of IEEE double precision floating-point number is
A) 23 B) 11
C) 8 D) 52
- The decimal equivalent of 2's complement number 11110010 is
A) +2 B) -2
C) -14 D) +14
- If the biased exponent of a IEEE single precision number is 127 then the actual exponent is
A) 0 B) 54
C) 127 D) 1
- On a processor with single-bus organization, the sequence of operations to add the contents of register R1 to those of R2 and store the result in R3 is
A) 1. R1out, Yin, R2out, SelectY
2. Add, Zin
3. Zout, R3in
C) 1. R1out, Yin, R2out, SelectY, Add, Zin
2. Zout, R3in
B) 1. R1out, Yin
2. R2out, SelectY, Add, Zin
3. Zout, R3in
D) 1. R1out, Yin, R2out
2. SelectY, Add, Zin
3. Zout, R3in
- On a processor with three-bus organization, the ALU may simply pass one of its 2 input operands unmodified to bus C. The control signals for such an operation is
A) $R=A$ or $R=B$ B) $A=R$ or $B=R$
C) Rout, Ain or Rout, Bin D) Aout, Rin or Bout, Rin

11. In microprogrammed control unit the control functions are implemented in
- A) main memory B) hardware
C) software D) secondary memory
12. control unit is flexible to accommodate new instructions.
- A) Single-bus B) Three-bus
C) Microprogrammed D) Hardwired
13. In memory cell, the information is stored in the form of charge on the capacitor
- A) Static RAM B) Dynamic RAM
C) secondary D) Read Only Memory
14. What is the storage capacity of 4096 x (512x8) cell array dynamic memory chip?
- A) 1 Mega bytes B) 2 Mega bytes
C) 4 Mega bytes D) 8 Mega bytes
15. The addresses of instruction and data used in a source program are referred to as
- A) logical addresses B) main memory addresses
C) physical addresses D) secondary memory addresses
16. The mapping function [(Main Memory Block address) MOD (Number of blocks in cache)] is used by:
- A) Set associative cache B) Direct mapped cache
C) Associative mapped cache D) Indirect mapped cache
17. The method which offers higher speeds of I/O transfers is
- A) Program-controlled I/O B) I/O mapped I/O
C) Interrupts D) DMA
18. The method of synchronizing the processor with the I/O device in which the device sends a signal when it is ready is
- A) Bus arbitration B) Cycle stealing
C) Exception D) Interrupt
19. The bus standard, PCI is the acronym for
- A) Personal Computer Interconnect B) Personal Computer Interface
C) Peripheral Component Interconnect D) Peripheral Component Interface
20. The method of accessing the I/O devices by repeatedly checking the status flags is
- A) Program-controlled I/O B) I/O mapped I/O
C) Memory-mapped I/O D) Interrupt driven I/O

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Show the connection between processor and the memory with suitable diagram and explain the role of different processor registers.	6	L*1	1	1
	b) Explain the relative addressing mode with proper syntax and suitable example.	5	L2	1	1
	c) Perform $14 \div 3$ using non-restoring division method.	5	L3	2	1,2
2.	a) Perform bit-pair recoded multiplication of -15 (multiplicand) and +7 (multiplier).	6	L3	2	1,2
	b) Draw the block diagram of sequential circuit multiplier and list the steps involved in multiplying two unsigned numbers.	5	L2	2	1,2
	c) Perform $21 \div 5$ using restoring division method.	5	L3	2	1,2
3.	a) Multiply 55 (multiplicand) and 43 (multiplier) using carry-save addition of summands method.	6	L3	2	1,2
	b) Using the branching concept, write an assembly program to add n numbers stored in contiguous locations in the memory.	5	L2	1	1
	c) Represent $+1.0110 \times 2^6$ in IEEE single-precision floating-point number format.	5	L3	2	1,2
Unit – II					
4.	a) Describe the internal organization of a 16 X 8 memory chip with neat block diagram.	6	L2	4	1,2,3

b)	Write the control steps required to execute a complete instruction <i>ADD R3, R2, R1</i> in a three-bus organization. (Note: This instruction adds the contents of register R3 to those of R2 and stores the result in R1).	5	L3	3	1,2
c)	With the block diagram describe the single-bus organization of processor data path.	5	L2	3	1,2
5. a)	Write the control sequence for the execution of conditional branch instruction in a single-bus organization.	6	L3	3	1,2
b)	Explain the basic organization of a hardwired control unit.	5	L2	3	1,2
c)	A set associative cache consists of a total of 128 blocks divided into sets of 2 blocks each. The main memory contains 4096 blocks, each consisting of 16 words. What is the number of bits in each of the TAG, SET and WORD fields of the address?	5	L3	4	1,2,3
6. a)	With necessary block diagram explain the process of virtual memory address translation.	6	L2	4	1,2,3
b)	Write the control steps required to execute a complete instruction <i>Add (R1), R2</i> in a single-bus organization.	5	L3	3	1,2
c)	What is ROM? Explain the structure of a ROM cell with a neat diagram.	5	L1	4	1,2,3

Unit – III

7. a)	With a neat diagram of registers used in DMA (Direct Memory Access) interface, explain the process of DMA transfer.	6	L2	5	1
b)	Explain the mechanism used to handle simultaneous interrupt requests from multiple devices through a single interrupt line.	5	L2	5	1
c)	Explain the tree structure of USB with split bus operation.	5	L2	5	1
8. a)	With a neat diagram, explain the centralized bus arbitration mechanism.	6	L2	5	1
b)	With neat timing diagram illustrate the synchronous bus data transfer during an input operation.	5	L2	5	1
c)	With suitable diagram explain the interrupt nesting scheme.	5	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
